

CodeAlpha – Sentiment Analysis using Social Media Data

Task 4: Sentiment Analysis

This project is completed as part of the CodeAlpha Data Analytics Internship. The objective of Task 4 is to analyze social media reviews and classify public opinion using sentiment analysis techniques. Twitter data is used in this task, strictly following the internship instructions.

Project Objective

The main objective of this task is to analyze social media text data and classify it into positive, negative, or neutral sentiments. This helps in understanding public opinion and sentiment trends on social media platforms.

Dataset Information

The dataset used for this task is a Twitter sentiment dataset obtained from Kaggle. The dataset contains tweet text collected from social media. Each row represents a tweet, and the analysis is performed on the text column. This dataset fully satisfies the requirement of using social media reviews as mentioned in the task instructions.

Tools and Technologies Used

Google Colab is used as the development environment. Python is used as the programming language. Pandas is used for data handling and analysis. VADER Sentiment Analyzer is used for sentiment classification. Matplotlib is used for visualizing sentiment distribution.

Methodology

The Twitter dataset is first loaded into Google Colab. The tweet text is then analyzed using natural language processing techniques. VADER sentiment analysis is applied to calculate sentiment polarity scores for each tweet. Based on the compound score, tweets are classified into positive, negative, or neutral sentiments. The results are summarized and visualized using a bar chart.

Sentiment Classification

Positive sentiment indicates tweets with favorable opinions.

Negative sentiment indicates tweets with unfavorable opinions.

Neutral sentiment indicates tweets with no strong emotional tone.

Visualization

A bar chart is created to display the distribution of positive, negative, and neutral tweets. This visualization helps in understanding overall public sentiment on social media.

Key Insights

The analysis provides an overview of how users express opinions on social media. It helps identify whether the general sentiment is positive, negative, or neutral. Such insights can be useful for decision-making, marketing analysis, and understanding public perception.

Conclusion

This project successfully fulfills Task 4 of the CodeAlpha Data Analytics Internship. By using a Twitter-based social media dataset and applying sentiment analysis techniques in Google Colab, meaningful insights into public opinion are obtained in a clear and structured manner.

